EXPERIMENT 19

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

Logistic Regression

Aim: Predict customer purchase using Logistic Regression.

Python Program

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
df=pd.read_csv('Social_Network_Ads.csv')
x=df[['Age','EstimatedSalary']]
y=df['Purchased']
xtr,xte,ytr,yte=train_test_split(x,y,test_size=0.2,random_state=1)
model=LogisticRegression(max_iter=1000)
model.fit(xtr,ytr)
print("Accuracy:",model.score(xte,yte))
print("Prediction:",model.predict(xte[:5]))
```

Result

The logistic regression model predicts purchase class and displays test accuracy.

Note: For CSV programs, keep the named CSV file in the same folder and check that column names match.